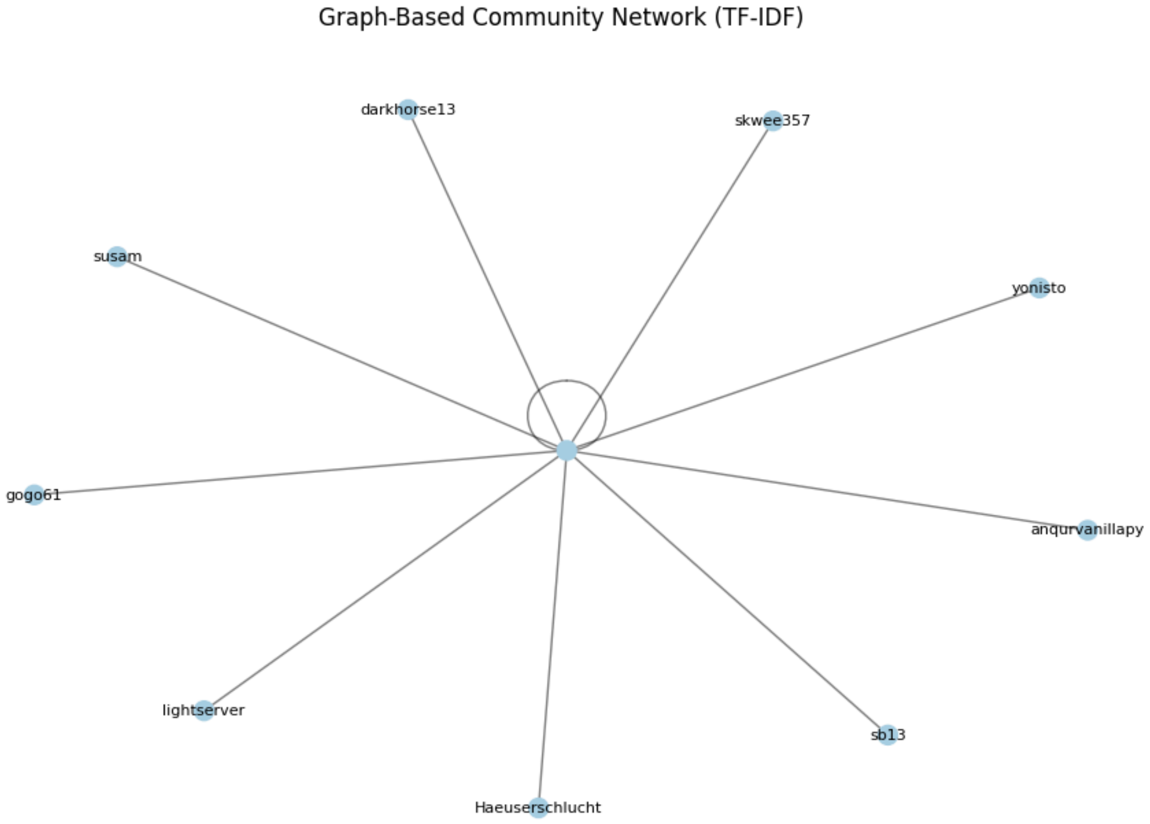
Develop a structure based social media analytics model for any business. Build models for community detection and influence analysis using libraries like NetworkX, Gephi, pr Neo4j. Analyze social media graphs and connections using APIs like LinkedIn API or Twitter API v2

```
import pandas as pd

df_posts = pd.read_csv('/content/drive/MyDrive/SMA_DATASET/merged_questions.csv')
df_comments = pd.read_csv('/content/drive/MyDrive/SMA_DATASET/merged_comments.csv')
```

```
import networkx as nx
import community.community_louvain as louvain
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import matplotlib.pyplot as plt

posts_info = df_posts.groupby('User')['Question'].apply(' '.join).reset_index(name='Posts
comments_info = df_comments.groupby('Author')['Comment'].apply(' '.join).reset_index(name
user_activity = pd.merge(posts_info, comments_info, left_on='User', right_on='Author', hc
tfidf_vectorizer = TfidfVectorizer(stop_words='english')
tfidf_matrix = tfidf_vectorizer.fit_transform(user_activity['PostsText'] + ' ' + user_act
cosine_similarities = cosine_similarity(tfidf_matrix, tfidf_matrix)
G = nx.Graph()
for i in range(len(user_activity)):
    for j in range(i + 1, len(user_activity)):
        similarity = cosine_similarities[i, j]
        if similarity > 0.2:
            G.add_edge(user_activity['User'][i], user_activity['User'][j])

partition = louvain.best_partition(G)
user_activity['Community'] = user_activity['User'].map(partition)
pos = nx.spring_layout(G)
plt.figure(figsize=(12, 8))

# Draw nodes colored by community
colors = [partition[node] for node in G.nodes]
nx.draw_networkx_nodes(G, pos, node_color=colors, cmap=plt.cm.Paired, node_size=100)
nx.draw_networkx_edges(G, pos, alpha=0.5)
labels = {node: user_activity.loc[user_activity['User'] == node, 'User'].values[0] for nc
nx.draw_networkx_labels(G, pos, labels, font_size=8)

plt.title('Graph-Based Community Network (TF-IDF)')
plt.axis('off')
plt.show()
```




```
user_activity
```

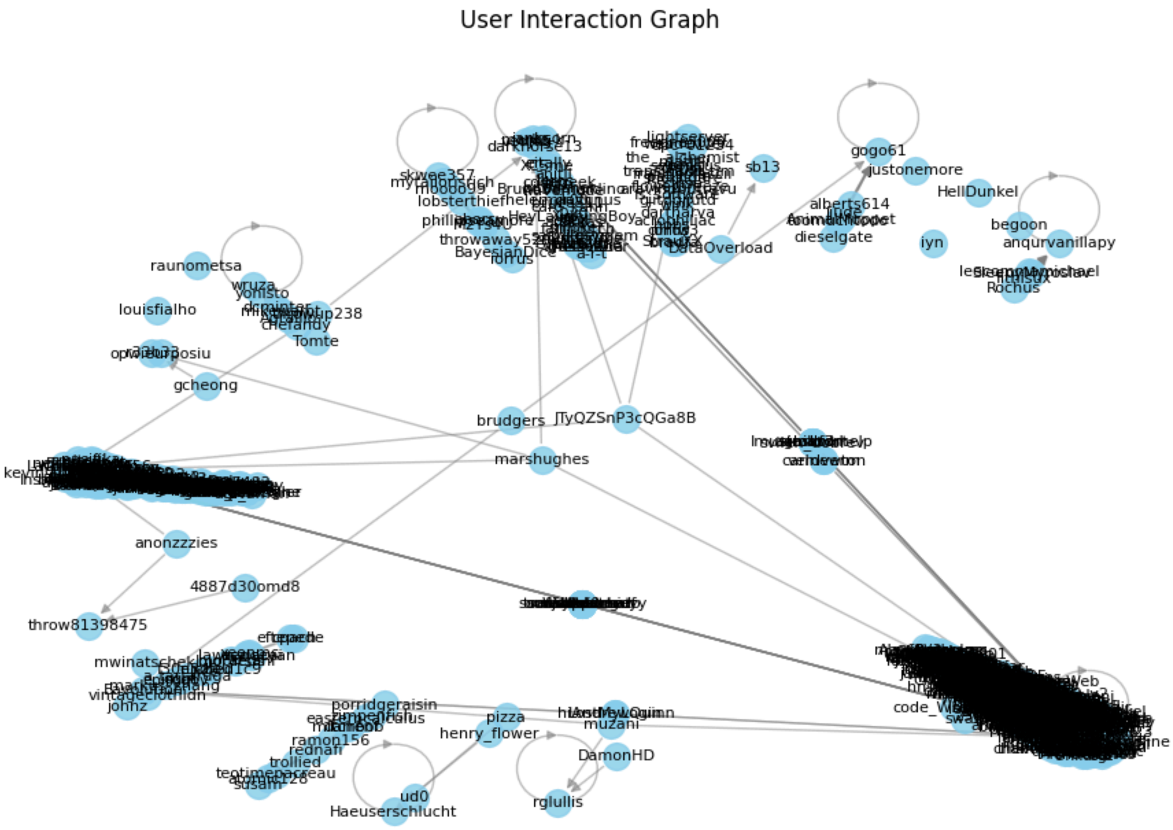
	User	PostsText	Author	CommentsText	Community
0			0100101101001	I'm working on a browser extension that aims t...	0.0
1			01jonny01	I'd rather put up with the speeding than aaski...	0.0
2			101008	A Django UI Desktop app, a là Docker Desktop, ...	0.0
3			147	I'm in the early stages of working on a Kubern...	0.0
4			234120987654	Fine-tuning an LLM to create long-form podcast...	0.0
...	...	...	...	...	...
789			zciwor	I'm working on cataloging open source hardware...	0.0
790			zer0x4d	An ode to DreamweaverDreamweaver was how I lea...	0.0

```
comments_per_link = df_comments.groupby('Link')['Comment'].count().reset_index()
df_combined = pd.merge(df_posts, comments_per_link, on='Link', how='outer').fillna(0)
df_combined['InfluenceScore'] = df_combined['Score'] + df_combined['Comment']
most_influential_user = df_combined.groupby('User')['InfluenceScore'].sum().idxmax()
print(df_combined.head(2))
print()
print(f"The most influential user is: {most_influential_user}")
```

```
                                              Title          User  \
0  Ask HN: What fiction/non-fiction book should e...    Eavolution
1  Ask HN: Can I really create a company around m...  darkhorse13

  Relative time        Date      Time  Score  \
0    7 days ago  2025-02-18  00:40:33     35
1    4 days ago  2025-02-18  08:37:01     73

                                            Link  \
0  https://news.ycombinator.com/item?id=43084739
1  https://news.ycombinator.com/item?id=43087383

                                           Question  Comment  InfluenceScore
0  This should be interpreted in the broadest sen...     50.0            85.0
1  This is a serious question because I'll be the...    124.0           197.0

The most influential user is: david927
```

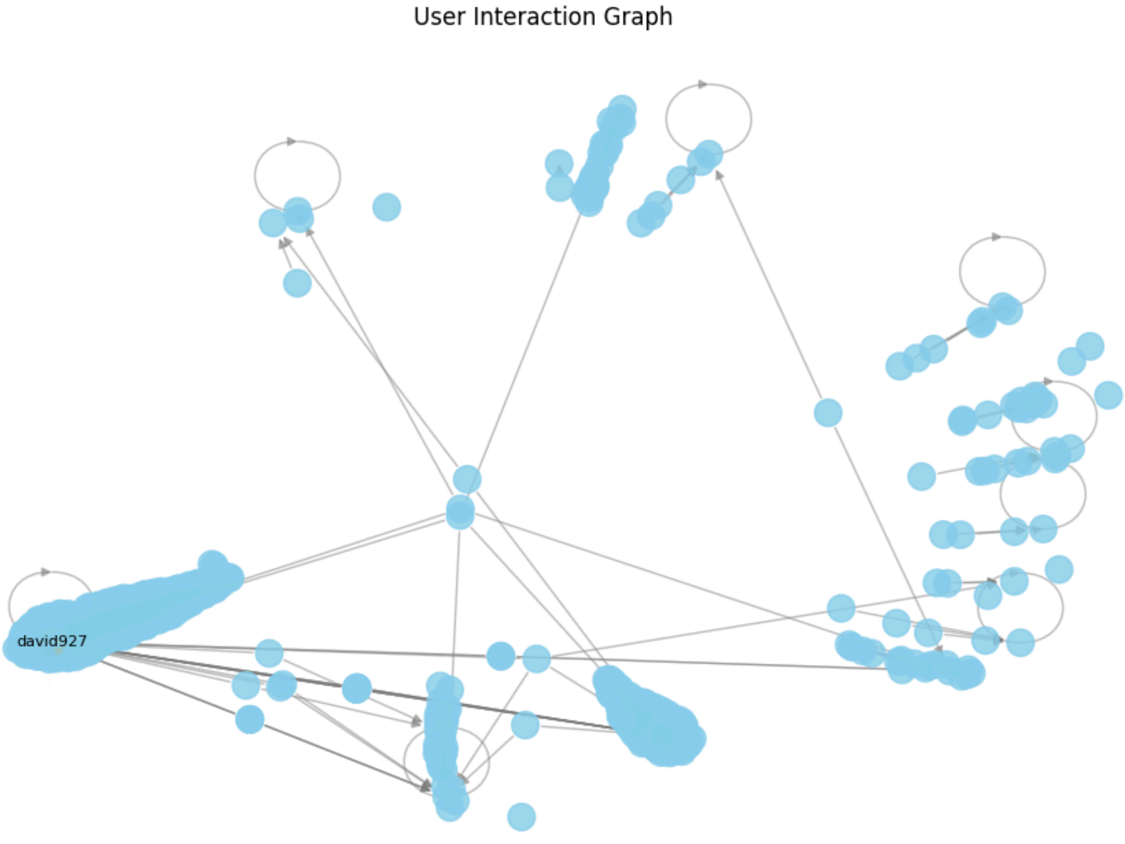
```
G = nx.DiGraph()

users = set(df_posts['User']).union(set(df_comments['Author']))
G.add_nodes_from(users)

for index, row in df_comments.iterrows():
```

```
    G.add_edge(row['Author'], df_posts.loc[df_posts['Link'] == row['Link'], 'User'].value

pos = nx.spring_layout(G)
plt.figure(figsize=(12, 8))

nx.draw_networkx_nodes(G, pos, node_size=200, node_color='skyblue', alpha=0.8)
nx.draw_networkx_edges(G, pos, width=1.0, alpha=0.5, edge_color='gray')
nx.draw_networkx_labels(G, pos, font_size=8)

plt.title('User Interaction Graph')
plt.axis('off')
plt.show()
```




```
G = nx.DiGraph()

users = set(df_posts['User']).union(set(df_comments['Author']))
G.add_nodes_from(users)
```

```
for index, row in df_comments.iterrows():
    G.add_edge(row['Author'], df_posts.loc[df_posts['Link'] == row['Link'], 'User'].value

max_in_degree_node = max(dict(G.in_degree()).items(), key=lambda x: x[1])[0]

pos = nx.spring_layout(G)
plt.figure(figsize=(12, 8))

node_colors = ['orange' if node == max_in_degree_node else 'skyblue' for node in G.nodes]
nx.draw_networkx_nodes(G, pos, node_size=200, node_color=node_colors, alpha=0.8)
nx.draw_networkx_edges(G, pos, width=1.0, alpha=0.5, edge_color='gray')

labels = {max_in_degree_node: max_in_degree_node}
nx.draw_networkx_labels(G, pos, labels, font_size=8)

plt.title('User Interaction Graph')
plt.axis('off')
plt.show()
```



Centrality Metrics for Influence Analysis

```
pagerank_scores = nx.pagerank(G)
most_influential = max(pagerank_scores, key=pagerank_scores.get)
print(f"Most influential user based on PageRank: {most_influential}")
```

```
Most influential user based on PageRank: david927
```

```
!pip install prophet
```

Show hidden output

Time-Series Trend Analysis

```
from prophet import Prophet

df_trend = df_posts[['Time', 'Score']]
df_trend.columns = ['ds', 'y']

model = Prophet()
model.fit(df_trend)

future = model.make_future_dataframe(periods=30)
forecast = model.predict(future)

model.plot(forecast)
```

```
/usr/local/lib/python3.11/dist-packages/prophet/forecaster.py:1133: UserWarning: Coul
  self.history_dates = pd.to_datetime(pd.Series(df['ds'].unique(), name='ds')).sort_v
/usr/local/lib/python3.11/dist-packages/prophet/forecaster.py:287: UserWarning: Coulc
  df['ds'] = pd.to_datetime(df['ds'])
INFO:prophet:Disabling yearly seasonality. Run prophet with yearly_seasonality=True t
INFO:prophet:Disabling weekly seasonality. Run prophet with weekly_seasonality=True t
INFO:prophet:Disabling daily seasonality. Run prophet with daily_seasonality=True to
INFO:prophet:n_changepoints greater than number of observations. Using 21.
DEBUG:cmdstanpy:input tempfile: /tmp/tmp1xgje9s6/ow36jv84.json
DEBUG:cmdstanpy:input tempfile: /tmp/tmp1xgje9s6/izu2f14d.json
DEBUG:cmdstanpy:idx 0
DEBUG:cmdstanpy:running CmdStan, num_threads: None
DEBUG:cmdstanpy:CmdStan args: ['/usr/local/lib/python3.11/dist-packages/prophet/stan_
05:35:25 - cmdstanpy - INFO - Chain [1] start processing
INFO:cmdstanpy:Chain [1] start processing
05:35:25 - cmdstanpy - INFO - Chain [1] done processing
INFO:cmdstanpy:Chain [1] done processing
```